Zebra Aurora Focus JavaScript Programmer's Guide

/docs/javascript-developers-guide

Zebra Aurora Focus JavaScript Programmer's Guide

This guide provides information on developing custom scripts for Zebra Aurora Focus using the integrated JavaScript editor. It includes details on the scripting environment, operation modes, debugging techniques, sample scripts, and full interface descriptions for handling inspection results.

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About this Guide

/docs/t-aurora-js-about-this-guide

About this Guide

This document provides information on using JavaScript within Zebra Aurora Focus, including debugging techniques, sample scripts, and interface descriptions for barcode and anomaly detection results.

See Also

- Notational Conventions
- Icon Conventions
- Service Information

Notational Conventions

/docs/r-notational-conventions

Notational Conventions

The following notational conventions make the content of this document easy to navigate.

- **Bold** text is used to highlight the following:
  - Dialog box, window, and screen names
  - Dropdown list and list box names
  - Checkbox and radio button names
  - Icons on a screen
  - Key names on a keypad
  - Button names on a screen
- Bullets (•) indicate:
  - Action items
  - List of alternatives
  - Lists of required steps that are not necessarily sequential
- Sequential lists (for example, those that describe step-by-step procedures) appear as numbered lists.

See Also

- About this Guide

Icon Conventions

/docs/r-icon-conventions

Icon Conventions

The documentation set is designed to give the reader more visual clues. The following visual indicators are used throughout the documentation set.

NOTE

The text here indicates information that is supplemental for the user to know and that is not required to complete a task.

INFO

The text here indicates information that is important for the user to know.

CAUTION

If the precaution is not heeded, the user could receive a minor or moderate injury.

WARNING

If danger is not avoided, the user CAN be seriously injured or killed.

DANGER

If danger is not avoided, the user WILL be seriously injured or killed.

See Also

- About this Guide

Service Information

/docs/c-service-information

Service Information

If you have a problem with your equipment, contact Zebra Global Customer Support for your region. Contact information is available at: zebra.com/support.

When contacting support, please have the following information available:

- Serial number of the unit
- Model number or product name
- Software/firmware type and version number

Zebra responds to calls by email, telephone, or fax within the time limits set forth in support agreements.

If your problem cannot be solved by Zebra Customer Support, you may need to return your equipment for servicing and will be given specific directions. Zebra is not responsible for any damages incurred during shipment if the approved shipping container is not used. Shipping the units improperly can possibly void the warranty.

If you purchased your Zebra business product from a Zebra business partner, contact that business partner for support.

See Also

- About this Guide

JavaScript Environment

/docs/c-aurora-js-environment

JavaScript Environment

JavaScript formatting provides the capability of post-processing inspection results, changing job status or FTP path, and sending manipulated results to different communication channels (HID, Serial, CDC, TCP).

Aurora Focus provides an execution environment based on the ECMAScript 2024 standard. For additional information, go to tc39.es/ecma262/2024/. Minor environmental restrictions are present.

Using the JavaScript features available in Aurora Focus is analogous to those available in web browsers such as Chrome or Edge and standalone JavaScript environments. In most cases, it is possible to rely on online tutorials and the help of AI assistants without any specific code customization.

The JavaScript runtime in Aurora Focus provides the following features:

- Just-in-Time Compilation - the script is optimized during execution, generating efficient machine code after several iterations, as if written in C or C++.
- Dedicated Scripting Environment - separation of the scripting environment from the vision runtime environment ensures that problems with the scripting environment do not cause significant problems with job execution. This ensures that scripting problems should not break the acquisition environment.

See Also

- Using the JavaScript Editor
- Using the Editor for a Deployed Job
- Limitations

Using the JavaScript Editor

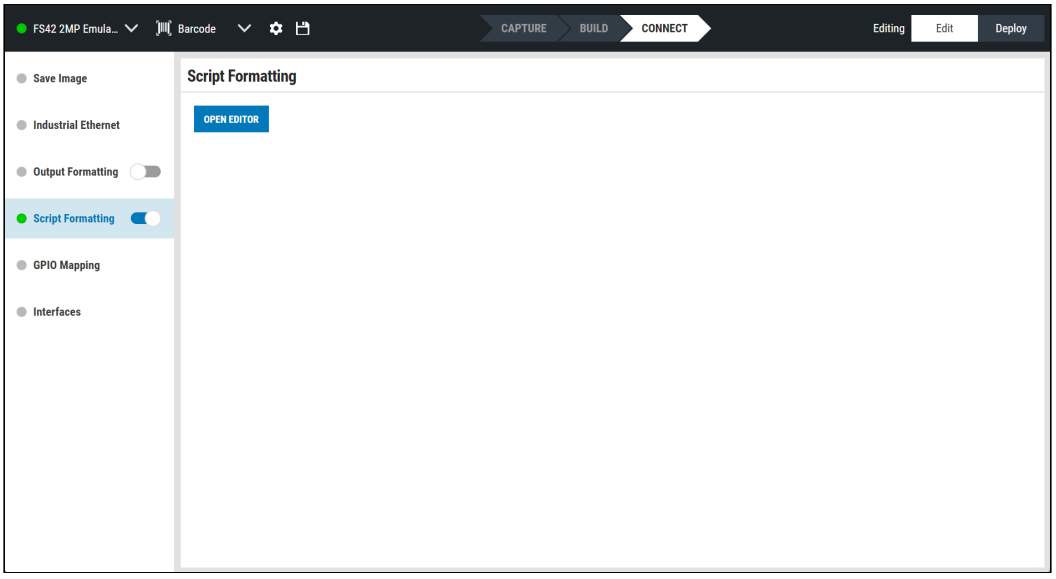
/docs/t-aurora-js-using-the-javascript-editor

Using the JavaScript Editor

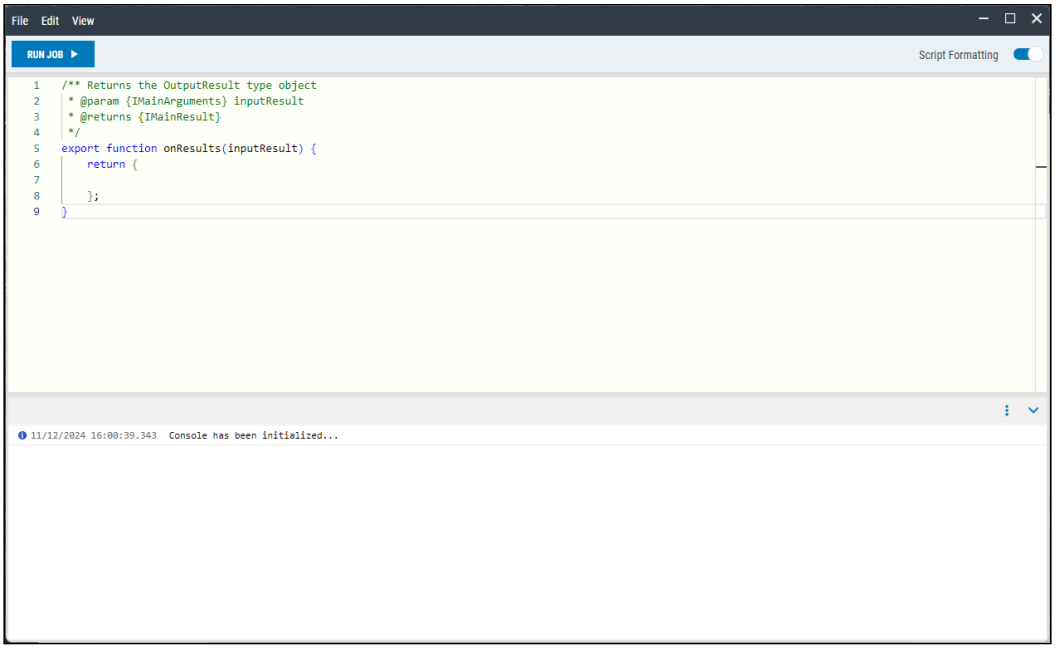
Access **Script Formatting** from the **Connect** tab to create custom scripts.

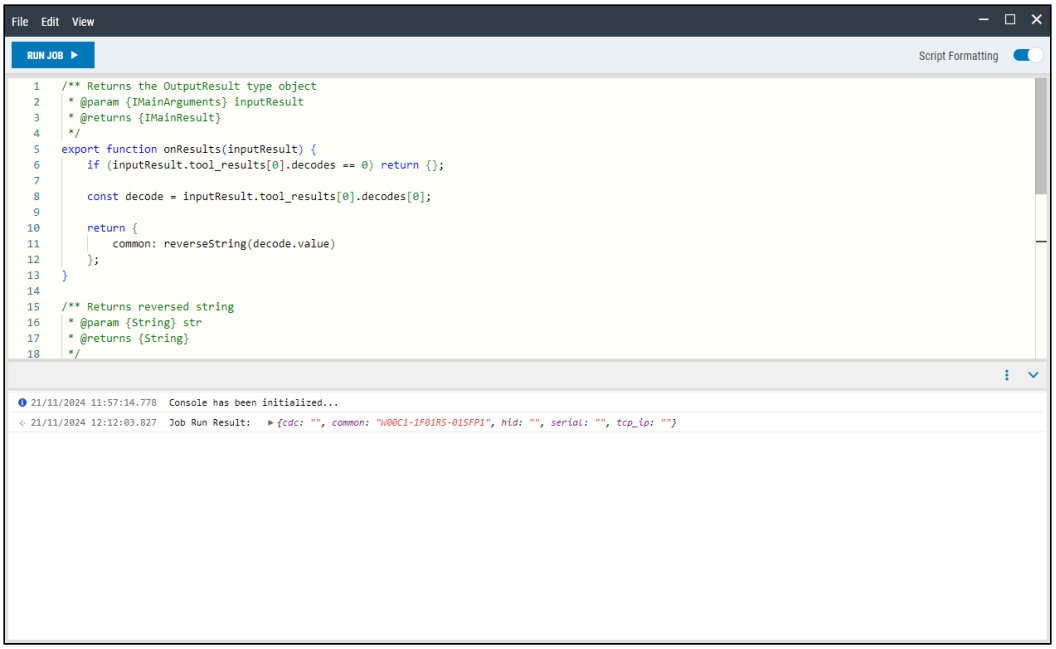
> **NOTE**
>
> JavaScript Formatting is supported on Aurora Focus version 9 and above.

1. Enable **Script Formatting** from the menu and click **Open Editor** to open a new editor window.

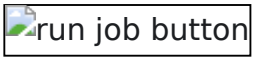


2. While in the editor, observe the UI elements: the navigation menu, the run job button, and the script formatting slider.

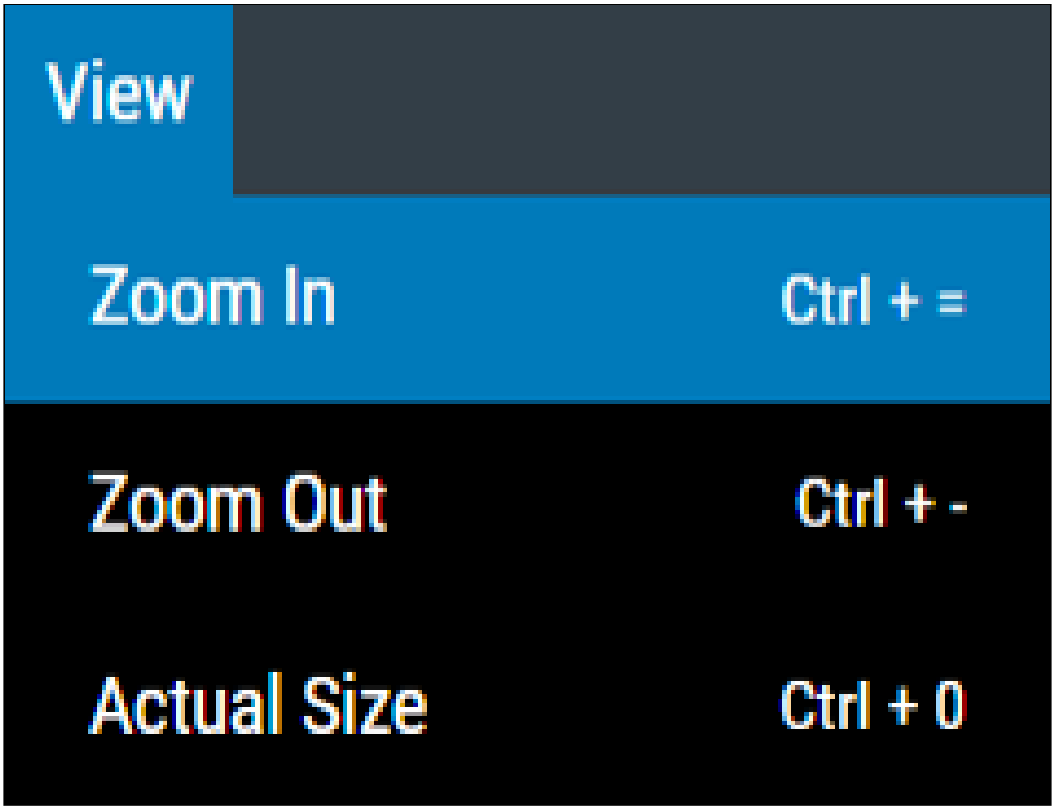

File Edit View
RUN JOB
Script Formatting
/** Returns the OutputResult type object
* @param {IMainArguments} inputResult
* @returns {IMainResult}
*/
export function onResults(inputResult) {
return {
};
}
11/12/2024 16:00:39.343 Console has been initialized...


File Edit View
RUN JOB
Script Formatting
/** Returns the OutputResult type object
* @param {IMainArguments} inputResult
* @returns {IMainResult}
*/
export function onResults(inputResult) {
if (inputResult.tool_results[0].decodes == 0) return {};
const decode = inputResult.tool_results[0].decodes[0];
return {
common: reverseString(decode.value)
};
}
/** Returns reversed string
* @param {String} str
* @returns {String}
*/
21/11/2024 11:57:14.778 Console has been initialized...
21/11/2024 12:12:03.827 Job Run Result: {cdc: "", common: "W00C1-1F01RS-01SFP1", hid: "", serial: "", tcp_ip: ""}

File
Edit
View


run job button

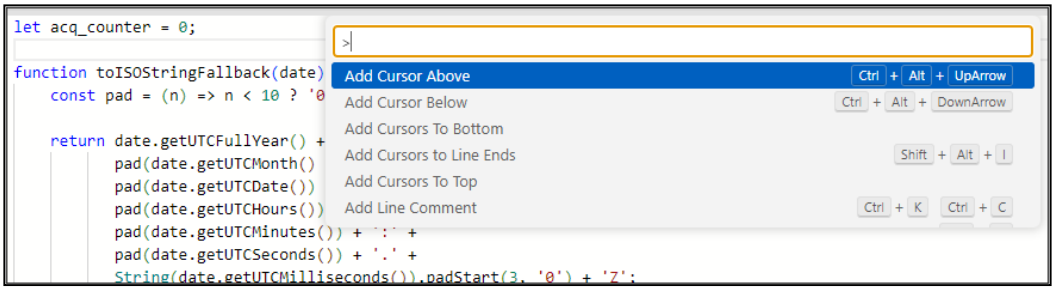
Script Formatting



JavaScript Editor UI Elements

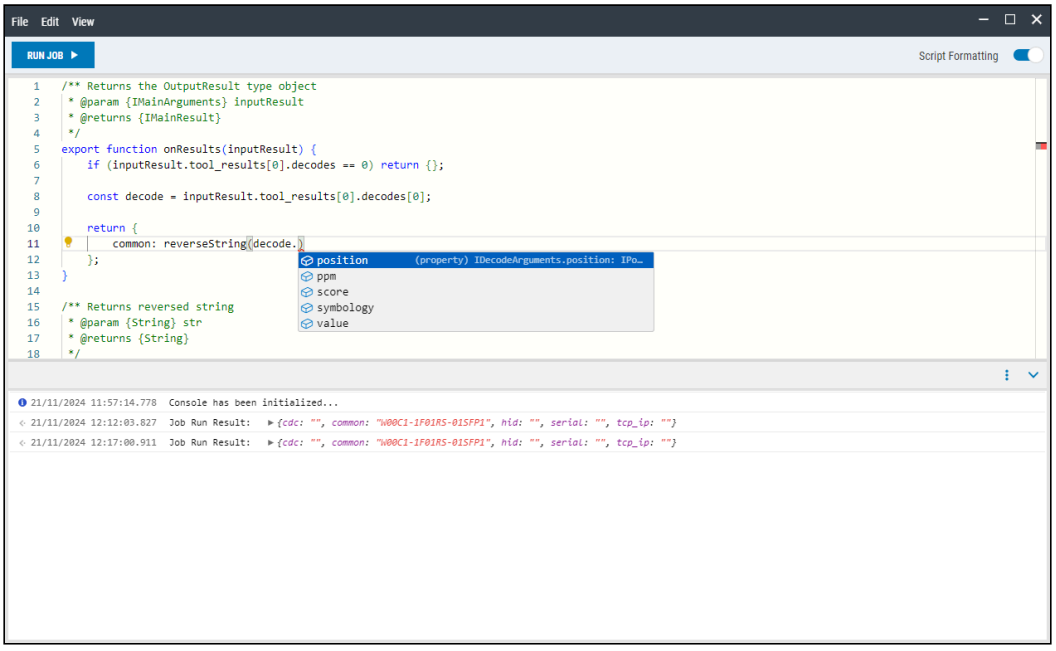
UI Element	Description
Navigation Menu	The top navigation menu provides access to **File**, **Edit**, and **View** functions for the application. • Open an existing file with JavaScript code and save JavaScript to an external file from the **File** menu. JavaScript, by default, is saved in a job file when saving a whole job or deploying a job to the device. • Use templates and snippets to quickly develop new JavaScript code that is specific to your use case from the **Edit** menu. • Adjust the editor font size from the **View** menu.
Run Job Button	Click **Run Job** on the control bar to validate test code in the editor by running a job on an image in the Filmstrip in Aurora Focus. You can also trigger a job by pressing **F5** while in the editor.
Script Formatting	Disable JavaScript formatting by using the **Script Formatting** toggle in the top-right corner of the editor window.

3. Press **F1** while inside the editor for a list of all functionalities.



4. Use IntelliSense for script recommendations to enable code completion. When a function has JSDoc documentation, the editor assumes its variable types and provides a recommendation for object properties. IntelliSense is also useful for standard JavaScript interfaces such as `JSON` or `Date`.

> ⓘ **NOTE**
>
> It is recommended to keep JSDoc comments for the `onResult` function. This is necessary to receive recommendations for input and output objects.



5. Monaco Editor powers the code editor to integrate functionalities such as adjusting the format, undo/redo, renaming symbols, and find/replace. The following code snippet is a basic script that copies the first detected barcode from the first tool to the output.

```
/** Returns the OutputResult type object
* @param {InputResult} inputResult
* @returns {OutputResult}
*/
export function onResults(inputResult) {
  return {
      common: inputResult.tool_results.decodes.value
  };
}
```

6. Use the console below the code editor to view logs. The log icon and text color change depending on the log category.

   - Use ⋮ to save the console logs to a file or clear the console.
   - Use ⌄ to minimize the console.

See Also

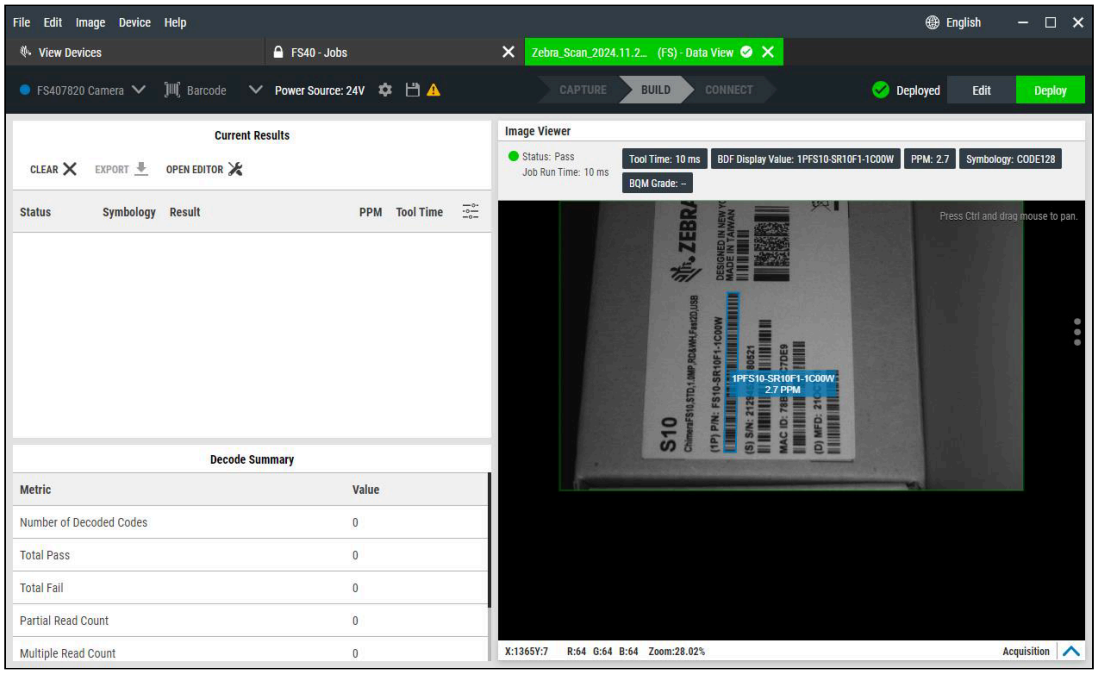
- JavaScript Environment

Using the Editor for a Deployed Job

/docs/t-aurora-js-using-the-editor-for-a-deployed-job

Using the Editor for a Deployed Job

Click **Open Editor** in the **Current Results** section while running a job in Aurora Focus to run the JavaScript editor.



See Also

- JavaScript Environment

Limitations

/docs/t-aurora-js-limitations

Limitations

Understand the current limitations of the script editor and strategize so you do not encounter any issues.

- The ability to use external modules using the `import` keyword and the required functions is disabled.
- Asynchronous functions are not supported. Helper functions necessary to implement asynchronous functions, such as `setTimeout`, are not present.
- Network or file system IO functions are not supported.
- If memory usage exceeds a specific limit, the scripting environment restarts (including restarting global variables). This is no different from other runtime errors, except for refreshing the state and the time it takes to restart the environment. This can occur if you create a global variable in an array and add elements in every iteration, but do not remove them.

See Also

- JavaScript Environment

Operation Mode

/docs/t-aurora-js-operation-mode

Operation Mode

Use the Aurora Focus JavaScript environment to handle input/output data, interact with hardware, and utilize job status logic.

See Also

- Program Structure
- Input Data
- Constant Data
- Output Data
- Job Status
- TCPIP, RS232, and USB CDC Serial
- HID Keyboard
- FTP Control
- GPIO Ports
- Beeper
- Script Invocation
- Example Script Lifecycle
- Managing Execution Time
- Text Encoding

Program Structure

/docs/c-aurora-js-program-structure

Program Structure

Every script must contain the function `onResults`. This function takes one argument as a data structure and returns the structure.

The `onResults` function must be marked as export as in the following example:

```
export function onResults(inputResult) {

}
```

Input Data

/docs/t-aurora-js-input-data

Input Data

The function `onResults` takes only one argument. This argument contains the output structure of all tools in the current job.

The `tool_results` and `tool_results_by_names` fields contain tools that display in the job but allow access to them differently. For example, `tool_results` is a list of tools where each index already has a defined tool type. In cases where the engine does not yet support the tools, the index takes the value `null`. The tools are sorted in the order they display in the job.

`tool_results_by_names` allows access to tools by their name, using dot notation, and support from IntelliSense, so `tool_results_by_names.my_tool` is the correct syntax. In `tool_results_by_names`, only currently supported tools display after their names. When FlowBuilder uses multiple tools, each tool is accessible by its friendly name.

> **NOTE**
>
> A friendly name might contain characters that are not valid as part of a JavaScript variable name. In this case, all invalid characters are replaced with underscore characters. In some instances, two tools might have the same JavaScript name (for example, friendly names were `tool!` and `tool@` and both got translated to `tool_`). In this case, the red triangle with an exclamation mark displays on the top right side of the editor (right before the script enabling switch). The warning sign explains which tools have a naming collision and which friendly names can be changed.

The descriptions of both fields are generated dynamically during job editing. As a result, changing the tool or its order generates IntelliSense errors at the script editing stage.

Only incoming data from the following tools is supported:

- Read Barcode
- Read DPM
- Read DPM & Barcode

- Datacode
- Anomaly Detection
- Deep Learning OCR
- Contrast
- Locate Edge
- Edge Detect
- Edge Count
- Gradient Full
- Gradient Horizontal
- Gradient Vertical

These tools share a common type and display in the argument in the same order as in Aurora Focus.

The following code snippet describes the input argument in TypeScript. In a typical development process, it is recommended to use the field subtitles feature in the editor.

```
interface IJobInformationArguments {
 name: string;
 start_time: string;
 acquisition_images_count: number;
 metrics: IMetricsArgument;
}
interface IMetricsArguments {
 trigger_counts: number;
 trigger_overruns: number;
 total_pass: number;
 total_fail: number;
 job_time_min: number;
 job_time_max: number;
 job_time_average: number;
}

The job_success field indicates if the job was completed successfully.
The script can override this value by using an output structure.

interface IMainArguments {
 tool_results: TOOL_RESULTS_TYPE[];
 tool_results_by_names: TOOL_RESULTS_BY_NAMES_TYPE;
 job_info: IJobInformationArguments;
 job_success: boolean;
}

The following code snippet describes definitions for additional types.
The IBarcodeResultArguments interface in TypeScript defines the structure
for barcode scanning results.
It includes the following properties:
time: number; This number indicates the time a tool or process takes,
such as image acquisition.

tool_time: number; similar to time, this is a number that might indicate
the time taken by a tool or process.

success: boolean; indicates whether a specific operation or process was
successful (true) or not (false).

decode_time: number; represents the time taken to decode a barcode.

decode_count: number; represents the count of successful decodes.

decode_totalcount: number; represents the total count of attempted
decodes, successful or not.

decodes: IDecodeArguments[]; an array of objects that conform to the
IDecodeArguments interface. Each element in this array represents a
decoded result, and IDecodeArguments would be another interface that
describes the structure of these decoded results.
```

```
unmatched_decodes: IDecodeArguments[]; as above, but codes that do not
match the pattern specified in the tool have been detected.

statistics: IStatisticsArguments; contains tools execution statistics
accumulated between operations.

interface IBarcodeResultArguments {
 time: number;
 tool_time: number;
 success: boolean;
 decode_time: number;
 decode_count: number;
 decode_totalcount: number;
 decodes: IDecodeArguments[];
}

The IPointArguements interface defines an object structure representing a
point in a two-dimensional space. It includes two properties: x and y,
both numbers.
x: number; expected to be a number representing the x-coordinate of a
point in a two-dimensional space.

y: number; a number representing the y-coordinate of the point.

interface IPointArguments {
 x: number;
 y: number;
}

The IQualityGradeArguements interface defines an object structure for
representing a quality grade.
grade_: number; a number representing the numerical value of the grade.

letter_grade: string; a string representing the letter equivalent of the
grade, such as A, B, or C.

interface IQualityGradeArguments {
 grade_: number;
 letter_grade: string;
}

The IDecodeBQMArguments interface in TypeScript defines the structure for
barcode quality metrics (BQM). Each property typically involves quality
grading using the IQualityGradeArguements interface, which includes a
numerical grade and a letter grade.
interface IDecodeBQMArguments {
 overall_grade: IQualityGradeArguments;
 decodability: IQualityGradeArguments | null;
 decode: IQualityGradeArguments;
 defects: IQualityGradeArguments | null;
```

```
 minimum_edge_contrast: IQualityGradeArguments | null;
 modulation: IQualityGradeArguments;
 symbol_contrast: IQualityGradeArguments;
 quiet_zone: IQualityGradeArguments | null;
 left_quiet_zone: IQualityGradeArguments | null;
 right_quiet_zone: IQualityGradeArguments | null;
 wide_to_narrow_ratio: IQualityGradeArguments | null;
 inter_char_gap_2_narrow: IQualityGradeArguments | null;
 max_pixel_value_in_extended_area: number | null;
 maximum_reflectance: IQualityGradeArguments | null;
 minimum_reflectance: IQualityGradeArguments | null;
 unused_error_correction: IQualityGradeArguments | null;
 fixed_pattern_damage: IQualityGradeArguments | null;
 axial_nonuniformity: IQualityGradeArguments | null;
 grid_nonuniformity: IQualityGradeArguments | null;
 contrast_uniformity: IQualityGradeArguments | null;
 reflectance_margin: IQualityGradeArguments | null;
 horizontal_print_growth: IQualityGradeArguments | null;
 vertical_print_growth: IQualityGradeArguments | null;
}

The IStatisticsArguments interface defines the tool execution statistics,
considering all iterations (including the current iteration).
total_pass : number; total number of tool successes.

total_fail : number; the total number of tool failures.

tool_time_min : number; minimum tool-making time.

tool_time_max : number; maximum tool-making time.

tool_time_average : number; average execution time.

interface IStatisticsArguments {
total_pass: number;
total_fail: number;
tool_time_min: number;
tool_time_max: number;
tool_time_average: number;
}

The IDecodeArguments interface defines the structure of an object
representing the result og a decoding process, such as decoding a
barcode.
value: string; represents the decoded value, such as the data extracted
from a barcode.

raw_value: number[]; represents the decoded value, such as the data
extracted from a barcode in ASCII codes.

symbology: string; indicates the type or format of the decoded code, such
```

```
as QR or Code128.

position: IPointArguments[]; an array of IPointArguments objects, each
representing a point in two-dimensional space. It typically denotes the
positions or corners of the decoded symbol within an image or a scanning
field.

ppm: number | null; represents pixel per module, which is used to measure
barcode module density in the images being decoded. It can also be null,
indicating that this information is not available or applicable.

score: IDecodeBQMArguments | null; an object conforming to the
IDecodeBQMArguments interface, representing the quality score of the
decoding process. It can also be null, indicating that a score is not
available or applicable.

interface IDecodeArguments {
 value: string;
 raw_value: number[];
 symbology: string;
 position: IPointArguments[];
 ppm: number | null;
 score: IDecodeBQMArguments | null;
}

The IAnomalyDetectionResultArguments interface in TypeScript defines the
structure for anomaly detection results. It includes the following
properties:

time: number; a number that might indicate the time a tool or process
takes, such as image acquisition.

tool_time: number; similar to time, this number might indicate the time
taken by a tool or process.

success: boolean; indicates whether a specific operation or process was
successful (true) or not (false).

statistics: IStatisticsArguments; contains tools execution statistics
accumulated between operations.

valid: boolean; indicates whether an image is considered valid (true) or
not (false).

score: number; represents the score of an image.

confident: boolean; indicates whether a result is considered confident
(true) or not (false).

roi: IPointArguments[]; an array of IPointArguments objects, each
representing a point in two-dimensional space. It typically denotes the
```

```
positions or corners of the tool region of interest.

interface IAnomalyDetectionResultArguments {
 time: number;
 tool_time: number;
 success: boolean;
 statistics: IStatisticsArguments;
 valid: boolean;
 score: number;
 confident: boolean;
 roi: IPointArguments[];
}

The IDeepLearningOcrResultArguments interface in TypeScript defines the
structure for deep learning-based OCR results. It includes the following
properties:

time: number; a number that might indicate the time a tool or process
takes, such as image acquisition.

tool_time: number; similar to time, this number might indicate the time
taken by a tool or process.

success: boolean; indicates whether a specific operation or process was
successful (true) or not (false).

characters: ICharacterResultArguments[]; an array of
ICharacterResultArguments objects, each representing a decoded character.

ocr_result: string; represents the decoded characters.

statistics: IStatisticsArguments; contains tools execution statistics
accumulated between operations.

interface IDeepLearningOcrResultArguments {
 time: number;
 tool_time: number;
 success: boolean;
 characters: ICharacterResultArguments[];
 ocr_result: string;
 statistics: IStatisticsArguments;
}

The ICharacterResultArguments interface defines an object structure
representing a decoded character. It includes the following properties:

value: string; represents the confidence character.

roi: IPointArguments[]; an array of IPointArguments objects, each
representing a point in a two-dimensional space. It typically denotes the
positions or corners of the decoded character.
```

```
confidence: number; represents the confidence of the character.

quality: number; represents the quality of the character.

height: number; represents the height of the character.

is_part_of_result_string: boolean; indicates if a character is contained
in the result string.

interface ICharacterResultArguments {
    value: string;
    roi: IPointArguments[];
    confidence: number;
    quality: number;
    height: number;
    is_part_of_result_string: boolean;
}

The IContrastResultArguments interface in TypeScript defines the
structure for contrast results. It includes the following properties:

time: number; this number indicates the time taken by a tool or process
including image acquisition.

tool_time: number; similar to time, this number indicates the time taken
by a tool or process.

success: boolean; indicates whether a specific operation or process was
successful (true) or not (false).

value: number; indicates contrast value.

minimum_threshold: number; indicates the minimum threshold value set in
the tool configuration.

maximum_threshold: number; indicates the maximum threshold value set in
the tool configuration.

minimum_calculation_value: number; indicates the minimum pixel value.

maximum_calculation_value: number; indicates the maximum pixel value.

statistics: IStatisticsArguments; contains tools execution statistics
accumulated between operations.

interface IContrastResultArguments {
 time: number;
 tool_time: number;
 success: boolean;
 value: number;
```

```
 minimum_threshold: number;
 maximum_threshold: number;
 minimum_calculation_value: number;
 maximum_calculation_value: number;
 statistics: IStatisticsArguments;
}

The IEdgeResultArguments interface in TypeScript defines the structure
for edge detection results. It includes the following properties:

time: number; this number indicates the time taken by a tool or process
including image acquisition.

tool_time: number; similar to time, this number indicates the time taken
by a tool or process.

success: boolean; indicates whether a specific operation or process was
successful (true) or not (false).

edge_count: number; indicates the number of detected edges.

line_segments: ILineArguments[]; an array of ILineArguments objects, each
representing a line in two-dimensional space.

statistics: IStatisticsArguments; contains tools execution statistics
accumulated between operations.

interface IEdgeResultArguments {
 time: number;
 tool_time: number;
 success: boolean;
 edge_count: number;
 line_segments: ILineArguments[];
 statistics: IStatisticsArguments;
}

The ILineArguments interface defines an object structure representing a
line in a two-dimensional space. It includes two properties: point1 and
point2, both of which are IPointArguments.

point1: IPointArguments; represents the line start point.

point2: IPointArguments; represents the line endpoint.

interface ILineArguments {
 point1: IPointArguments;
 point2: IPointArguments;
}
```

Monaco Sandbox

```
export function onResults(input_result) {
  return {
      tcp_ip: "Input Data sandbox"
  };
}
```

Constant Data

/docs/t-aurora-js-constant-data

Constant Data

Each script has a global variable called `DEVICE` that is available from anywhere in the code. They contain constant data about the current device. When using an emulator, this variable contains applicable messages about a lack of support.

`DEVICE` is described by the following type:

```
interface IDevice {
 hostname: string;
 model: string;
 part_number: string;
 serial_number: string;
 uuid: string;
 firmware_version: string;
 type: DeviceType;
 sensor: ISensor;
 gpio: IGPIO;
}

Using the following types:

enum DeviceType {
  REAL = "REAL",
  EMULATED = "EMULATED",
}

enum GpioPortState {
  DISABLE = "DISABLE",
  TRIGGER = "TRIGGER",
  JOB_SWITCH = "JOB_SWITCH",
  ILLUMINATION_STROBE = "ILLUMINATION_STROBE",
  INPUT = "INPUT",
  OUTPUT_PUSH_PULL = "OUTPUT_PUSH_PULL",
  OUTPUT_HIGH_SIDE = "OUTPUT_HIGH_SIDE",
  OUTPUT_STATIC_HIGH = "OUTPUT_STATIC_HIGH",
  OUTPUT_STATIC_LOW = "OUTPUT_STATIC_LOW",
  OUTPUT = "OUTPUT",
}

interface IGPIOPort {
  number: number;
  state: IGpioPortState;
}

interface IGPIO {
  count: number;
  ports: IGPIOPort[];
}

interface ISensor {
  sensor_type: string;
  width: number;
  height: number;
}
```

Monaco Sandbox

```
export function onResults(input_result) {
return {
  tcp_ip: DEVICE.model
};
}
```

Output Data

/docs/t-aurora-js-output-data

Output Data

Control the device's output function by using `onResults` to return a specific and straightforward structure.

```
interface IFtpResult {
  common_file_name?: (string | null);
  json_file_name?: (string | null);
  img_file_name?: ((string | string[]) | null);
}

interface IMainResult {
  common?: (string | (number[] | Uint8ClampedArray | Uint8Array));
  tcp_ip?: (string | (number[] | Uint8ClampedArray | Uint8Array));
  serial?: (string | (number[] | Uint8ClampedArray | Uint8Array));
  hid?: (string | (IHIDCombinationResult | IHIDPauseResult | HIDKey |
string)[]);
  cdc?: (string | (number[] | Uint8ClampedArray | Uint8Array));
  ftp?: (IFtpResult | null);
  job_success?: (boolean | null);
}
```

Monaco Sandbox

```
export function onResults(input_result) {
return {
  common: "Output Data sandbox",
  job_success: true
};
}
```

Job Status

/docs/t-aurora-scripting-job-status

Job Status

The `job_success` property redefines the inspection Pass/Fail status based on a user-defined logic.

If assigned with `true` or `false` values, the inspection status is pass or fail. It does not change if the value is not specified.

> **NOTE**
>
> It is possible to access the inspection status with the `onResults` function argument. The property is named `job_success` as well.

```
/**
* Changes overall inspection status into Fail if the Read_Barcode_1
* The tool has not found a single CODE 128 barcode.
*
* Please do not remove 'export' keyword or this comment.
* onResults function triggers on the results of each job run.
* For more detailed information please refer to the documentation.
* @param {IMainArguments} input_result - object with results of the job.
* @returns {IMainResult} - object-defining outputs per interface.
*/
export function onResults(input_result) {
  const results = input_result.tool_results_by_names.Read_Barcode_1;
  if (results.decode_count != 1) {
     return {
         job_success: false
     }
  }
  if (results.decodes.symbology != "CODE128") {
     return {
         job_success: false
     }
  }
  return {
     job_success: input_result.job_success
  }
}
```

Monaco Sandbox

```
export function onResults(input_result) {
 return {
    job_success: input_result.job_success
 };
}
```

TCPIP, RS232, and USB CDC Serial

/docs/t-aurora-scripting-tcpip-rs323-and-usb-cdc-serial

TCPIP, RS232, and USB CDC Serial

Use the JavaScript feature to write data to TCP/IP, RS232, and USB CDC-Serial external interfaces.

```
common ? : (string | (number[] | Uint8ClampedArray | Uint8Array));
tcp_ip ? : (string | (number[] | Uint8ClampedArray | Uint8Array));
serial ? : (string | (number[] | Uint8ClampedArray | Uint8Array));
cdc ? : (string | (number[] | Uint8ClampedArray | Uint8Array));

The properties allow writing data to external interfaces: tcp_ip for
TCP/IP, serial for RS232, and cdc for USB CDC-Serial. For binary data,
assign either a string or an array of numbers for binary data. Numbers
must be within the 0-255 range.

If you assign a value to a specific property, it is sent with the
interface. If a value is assigned to the common property, it is also sent
with the interface. Otherwise, no value is sent.

:::note

Settings in the Communication tabs are respected by the script.

:::

/**
* Prints the string value to the TCP/IP interface, and the value
* enriched with prefixes and suffixes to the serial interface.
*
* Please do not remove the 'export' keyword or this comment.
* onResults function triggers on the results of each job run.
* For more detailed information, please refer to the documentation.
* @param {IMainArguments} input_result - object with results of the job.
* @returns {IMainResult} - object-defining outputs per interface.
*/
export function onResults(input_result) {
  const results = input_result.tool_results_by_names.Read_Barcode_1;
  const decode_text = results.decodes.value;
  const decode_with_prefix_and_suffix = [0x02,
...results.decodes.raw_value, 0x03];
  return {
    tcp_ip: decode_text,
    serial: decode_with_prefix_and_suffix
  }
}
```

Monaco Sandbox

```
export function onResults(input_result) {
return {
  tcp_ip: "TCP message"
```

```
    tcp_ip: "TCP message",
    serial: [0x02, 0x41, 0x42, 0x43, 0x03],
    cdc: "USB CDC message"
  };
  }
```

HID Keyboard

/docs/t-aurora-js-hid-keyboard

HID Keyboard

Enable the HID keyboard to use the device in HID keyboard mode. Assign a string, a single button, a key, and a modifier combination. You can also add a pause between keystrokes. All of these can be leveraged into a single assignment.

```
hid?: (string|(IHIDCombinationResult| IHIDPauseResult|HIDKey|string)[]);
```

An additional helper object, HID, is available to simplify HID usage. HID properties are described in the following table.

Helper Object HID Properties Property Description HID.key Gives access to a list of available keys. HID.combination Allows the creation of a key and a modifier combination. HID.pause Allows a pause between consecutive keystrokes to be created. HID.modifier Gives access to a list of available key modifiers.

> **NOTE**
>
> HID.pause argument must be between 0 and 25000 ms.
>
> HID.combination allows you to press, at the same time, a key and up to 8 different modifiers.
>
> HID.combination produces a series of consecutive keystrokes with the same modifiers when used with more than one key.
>
> Keystroke delay and preferred keyboard layout can be changed through the Communication tab in the device settings.
>
> If that property is not assigned, nothing is printed.

```
/**
* Opens notepad and types into it a decoded barcode value.
*
* Please do not remove the 'export' keyword or this comment.
* The onResults function triggers the results of each job run.
* For more detailed information, please refer to the documentation.
* @param {IMainArguments} input_result - object with results of the job.
* @returns {IMainResult} - object-defining outputs per interface.
*/
export function onResults(input_result) {
  const run_command = [HID.combination(HID.key.R,
HID.modifier.L_WINDOWS), HID.pause(200)]
  const type_notepad = ["notepad", HID.key.ENTER, HID.pause(1000)];
  const decode =
input_result.tool_results_by_names.Read_Barcode_1.decodes;
  return {
    hid: [...run_command, ...type_notepad, decode.value, HID.key.ENTER]
  };
}
```

FTP Control

/docs/t-aurora-js-ftp-control

FTP Control

The `ftp` field controls FTP output if enabled in the job. If the `ftp` field is not set, explicitly set to `undefined`, or set to `null`, then default behavior and names are used, as set in the FTP tab for the job.

If the `ftp` field is set to an object, the resulting file names and whether they are sent depend on the field's values.

If `img_file_name` is set to a single string, the resulting image names are set to that name with the image bank name and number appended to the end. If it is set to an array of strings, the resulting image names are set to the names of the strings (the first image takes the first string name, and so on).

If `common_file_name` is set, but `img_file_name` is not set, the name used with the image bank index is appended to the end. If neither is set, images are not sent over FTP.

If `json_file_name` is set, and the resulting JSON file name is set to that name. If `common_file_name` is set but `json_file_name` is not set, the common file name is used. If neither is set, JSON is not sent over FTP.

See Also

- Operation Mode

GPIO Ports

/docs/t-aurora-focus-js-gpio-ports

GPIO Ports

Use the `gpio` field to send a pulse to available ports. Parameters of the pulse, such as length and delay, can be specified in the **GPIO Mapping** menu in General Settings. Pulse polarity can be set through the script, using one of the three values of the helper `GpioPolarityResult` enum object.

```
gpio: IGpioPort[];

Helper Object GpioPolarityResult Properties
Value   Description
GpioPolarityResult.HIGH Pulse has a high value.
GpioPolarityResult.LOW  Pulse has a low value.
GpioPolarityResult.JOB_SETTINGS Pulse parameters are taken from the
specification in the GPIO Mapping settings in the job.
The field should be an array of objects. Each object contains two
properties: port_number and polarity. port_number is a zero-starting
number of the port. Polarity can be one of three values described in the
table above.

:::note

The initial value can be set through the GPIO Mapping tab in the job.

The port that is written must be set as an output. It can be done through
GPIO Mapping in device settings.

The GPIO field of the DEVICE object can be used.

:::

/**
 * Sends a pulse to the n-th port if the n-th tool execution was
successful.
 * @param {IMainArguments} input_result - object with results of the job.
 * @returns {IMainResult} - object defining outputs per interface.
 */
export function onResults(input_result) {
    // For each tool for the FlowBuilder
    // take the success value, if positive send HIGH pulse
    let gpio_results = [];
    for (let i = 0; i < input_result.tool_results.length; ++i) {
        if (DEVICE.gpio.ports[i].state === GpioPortState.OUTPUT) {
            if (input_result.tool_results[i].success) {
                gpio_results.push({
                    port_number: i,
                    polarity: GpioPolarityResult.JOB_SETTINGS
                })
            }
        }
    }
    return {
        gpio: gpio_results
    };
}
```

Monaco Sandbox

```
export function onResults(input_result) {
  return {
      gpio: [
          {
              port_number: 0,
              polarity: GpioPolarityResult.HIGH
          }
      ]
  };
}
```

Beeper

/docs/t-aurora-focus-js-beeper

Beeper

Use the beeper field to control the beeper of the device and the Zebra Integrated Multi-Function Light.

```
beeper?: IBeepBeeper | BeeperStatusResult;
```

Helper Object Beeper Properties Value Description Beeper.beep() This function triggers a beeper with a specified duration, tone, and volume. Helper objects, BeeperVolumeResult, BeeperToneResult, and BeeperDurationResult, can be used to specify these. Beeper.off() Does not trigger the beeper; it’s independent of job_success. Beeper.device_settings() Triggers the beeper with parameters specified in the General Settings tab in Aurora Focus.

> **NOTE**
> The beeper must be enabled using the General Settings tab in Aurora Focus before being used on the device.

```
/**
 * Beep if the overall grade is not good enough.
 * @param {IMainArguments} input_result - object with results of the job.
 * @returns {IMainResult} - object defining outputs per interface.
 */
export function onResults(input_result) {
    const decodes =
input_result.tool_results_by_names.Read_Barcode.decodes;
    const bad_decodes = decodes.filter(decode => decode.score === null ||
        decode.score.overall_grade.grade < 3)
    const bad_values = bad_decodes.map(decode => decode.value + ", grade:
" +
        decode.score?.overall_grade.grade);
    if (bad_values.length > 0) {
        console.error("Codes with bad grades: ", bad_values.join(" "))
        return {
            beeper: Beeper.beep(BeeperVolumeResult.HIGH,
BeeperToneResult.MEDIUM,
                BeeperDurationResult.MEDIUM)
        };
    } else {
        return {
            beeper: Beeper.off()
        };
    }
}
```

Monaco Sandbox

```
export function onResults(input_result) {
  return {
      beeper: Beeper.beep(
          BeeperVolumeResult.HIGH,
          BeeperToneResult.MEDIUM,
          BeeperDurationResult.MEDIUM
      )
  };
}
```

Script Invocation

/docs/t-aurora-js-script-invocation

Script Invocation

The script executes at the end of each job when results are available. Its execution time includes the job's execution time.

The `onResults` function executes new arguments at the end of each job iteration. If there are any errors in the script, a standard JavaScript exception is thrown and propagated to Aurora Focus with no outputs set.

> **NOTE**
>
> For more information, refer to Debugging.

During the first iteration, the script execution may take longer because script code not enclosed in functions is executed first, such as initializing global variables.

> **NOTE**
>
> All other forms of result formatting are disabled when JavaScript is enabled.

See Also

- Operation Mode

Example Script Lifecycle

/docs/c-aurora-js-example-script-lifecycle

Example Script Lifecycle

The lifecycle of the script includes four stages: deployment, two iterations, and edit mode.

In the deployment stage, the execution environment is prepared without actions. In the first iteration, global variables are initialized, and syntax errors are checked. The `onResults` function processes the `Test String`, updates counters, and logs results. The second iteration skips initialization and processes the `Wrong String`, adjusts counters, and logs accordingly. Finally, the script enters edit mode, unloading from memory, with variables set to reinitialize upon redeployment.

```
console.log("Script loading begin");
let no_match_counter = 0;
let match_counter = 0;
const match_string = "Test String";
let counter = 0;

/** Returns the OutputResult type object
* @param {InputResult} inputResult
* @returns {OutputResult}*/
export function onResults(inputResult) {
  console.log("onResult called count: ", counter);
  counter++; // select first decode for first tool
  if (inputResult.tools.decodes === match_string) {
      match_counter++;
  } else {
      no_match_counter++;
      console.error("No match string find!");
      return {
          common: "Match count: " + match_counter + " No match count: " +
no_match_counter
      };
  }
  console.log("Script loading end");
}
```

There are four main stages in the lifecycle of a script:

1. Deploy the job to initialize the script's execution environment.

2. Iteration 1:

    - The code outside the onResults function is executed. The global variables no_match_counter, match_counter, match_string, and counter are created and initialized. The console.log functions are executed. Any syntax errors in the script are reported at this stage, and its execution is performed if errors are detected.
    - The onResults function is executed. Assume that in this iteration, the read code contains a string, e.g., Test String.
    - Values are set, and logs are sent during this stage.
    - The following string is sent to all available communication channels: Match count: 1 No match count: 0.
    - Aurora Focus receives logs in the following order: Script loading begins, Script loading end, onResult called count: 0.
    - Global variables at the end of the script are as follows: no_match_counter = 0, match_counter = 1, counter = 1.
3. Iteration 2:

    - No initialization phase.
    - The onResults function is executed. Assume that in this iteration, the read code contains the string, e.g., Wrong String.
    - Values are set, and logs are sent during this stage.
    - The following string is sent to all available communication channels: Match count: 1 No match count: 1.
    - Aurora Focus receives logs in the following order: onResult called count: 0 and No match string find!. The second log belongs to a different category, as noted in the editor console.
    - Global variables at the end of the script are as follows: no_match_counter = 1, match_counter = 1, counter = 2.
4. Edit the job and unload the script from memory.

> ⓘ **NOTE**
>
> During the next deployment, the variables from Step 2a are re-initialized.

> ⓘ **NOTE**
>
> Each execution is always the first one when using an emulator or performing jobs on the camera while editing.

Monaco Sandbox

```
let counter = 0;

export function onResults(input_result) {
  counter += 1;
  return {
      tcp_ip: "Lifecycle run #" + counter
  };
}
```

Managing Execution Time

/docs/t-aurora-js-managing-execution-time

Managing Execution Time

The JavaScript execution time may be unstable at times. Avoid instability by understanding its sources, just-in-time compiler operation, and garbage collector usage.

While smaller scripts typically do not result in instability, observed irregularities in script execution times result from two primary sources:

- **Just-in-time compiler operation.** When developing a script, subsequent iterations after the initial script are accelerated.
- **The JavaScript memory model uses a garbage collector.** Scripts that use a lot of memory must periodically use the garbage collector to clean up unused memory during execution, reducing the time it takes to execute.

See Also

- Operation Mode

Text Encoding

/docs/c-aurora-js-text-encoding

Text Encoding

The read content of the barcode, data code, or other tools offering output as text is treated as a single-byte encoding in accordance with ISO-8859-1.

In the script, assume that you receive an array of bytes as they display within the string. Output data to communication channels is sent in the same form as presented in the script, byte by byte, without any encoding.

Use data visualization to display the output structure within the console object with ASCII encoding.

All non-printable characters are printed as `\\<CONTENT\\>`.

For example, the code `0x8F` in hexadecimal notation (not a valid ASCII character) is displayed as `\\<0x8F\\>`.

See Also

- Operation Mode

Debugging, Advanced Techniques, and Sample Scripts

/docs/t-aurora-js-sample-scripts-advanced-techniques-and-debugging

Debugging, Advanced Techniques, and Sample Scripts

This section includes debugging tactics and sample scripts to implement into your workflow.

See Also

- Debugging
- Using Simulated Data
- Using Custom Templates
- Obtaining the Device MAC Address
- Sample Scripts

Debugging

/docs/c-aurora-js-debugging

Debugging

After you have mounted the camera, you can use basic script debugging features. The device always sends script execution data when in deploy mode.

The editor has a console at the bottom that displays the following data:

The console displays the results returned by the `onResults` function. The results must match the expected field names of the structure. Data that does not match the schema is skipped.

Use of the console object display in the console.

Syntactic or runtime issues are reported to the console, as well as all JavaScript exceptions.

The object console provides the following methods for the logs category: `console.log`, `console.warn`, `console.error`, and `console.debug`. They work similarly to the `console.log` function found in environments such as web browsers. The function accepts any number of arguments that can be concatenated into a single string. The objects passed are converted to text strings according to the standard rules of the JavaScript language.

Chained instructions, multiple reassignments, nested loops, and other complex expressions might be challenging to understand without providing insight into how the data is transformed. The following code snippet does not sufficiently explain its function.

```
export function onResults(input_result) {
 if (input_result.tool_results.decodes.length === 0) { // do not process
when there are no decodes
     return {};
 }
 const orderedDecodes = input_result.tool_results.decodes.map(decode =>
decode.value).sort().join(',');
 return {
     tcp_ip: orderedDecodes,
 };
}
```

Decodes are transformed when the console object is valid. Split the steps to separate assignments and log the actual values on the console. This method tracks the progress of the data transformation step by step and quickly tracks bugs. The following example uses the console object.

```
export function onResults(input_result) {
 if (input_result.tool_results.decodes.length === 0) { // do not process
when there are no decodes
     return {};
 }
 const decodes = input_result.tool_results.decodes;
 const decodedValues = decodes.map(decode => decode.value);
 console.log('decoded values:', decodedValues);
 const sortedValues = decodedValues.sort();
 console.log('sorted values:', sortedValues);
 const joinedResult = sortedValues.join(',');
 console.log('joined result: ', joinedResult);
 return {
     tcp_ip: joinedResult,
 };
}
```

Each step is logged in the console, and each step of the decode transformation from the raw decodes to a single string is observable. Use different colors in the console. `log`, `warning` or `error` methods to improve readability.

> **NOTE**
>
> While it is helpful to debug this way, the first approach may be preferable in production code because it does not declare many variables and uses less memory. After you validate the code, writing a concise version is recommended. If it is essential to keep the debug version of the code, comment it out and keep it for future reference to understand its purpose.

```
export function onResults(input_result) {
 if (input_result.tool_results.decodes.length === 0) { // do not process when there are no decodes
     return {};
 }
 // map all decodes to a single string where all decodes are separated by a comma and ordered alphabetically
 const orderedDecodes = input_result.tool_results.decodes
     .map(decode => decode.value) // map array of decodes to array of docoded values
     .sort() // sort alphabetically
     .join(','); // join elements of the array with a comma to a single string
 return {
     tcp_ip: orderedDecodes // output thorugh tcp
 };
}
```

See Also

- Debugging, Advanced Techniques, and Sample Scripts

Monaco Sandbox

```
export function onResults(input_result) {
console.log("Debug sandbox execution");
return {
  tcp_ip: "Debug output"
};
}
```

Using Simulated Data

/docs/t-aurora-js-using-simulated-data

Using Simulated Data

If you cannot use real-world results, use simulated data until you can validate with representative data.

Assign the data to a variable to simulate an actual result of a camera read. The following example simulates the results of a ManyCode read.

```
export function onResults(input_result) {
   if (input_result.tool_results.decodes.length === 0) { // do not
process when there are no decodes
       return {};
   }

   // mock the data for development and testing purposes, replace it with
the actual data when possible
   const mockedData = ["D21OCT21", "78B8D65C7DE9", "S21294520180521",
"1PFS10-SR10F1-1C00W"];

   //const orderedDecodes = input_result.tool_results.decodes //
uncomment this line when we switch to actual data
   //.map(decode => decode.value)                                //
uncomment this line when we switch to actual data

   const orderedDecodes = mockedData // remove or comment out this line
when we switch to actual data
       .sort()
       .join(',');

   return {
       tcp_ip: orderedDecodes
   };
}
```

WARNING

Simulated data is hardcoded and user-defined. Use this method only for development and initial testing purposes. Conduct final tests using actual results to ensure the code works as intended.

Monaco Sandbox

```
export function onResults(input_result) {
  const mockedData = ["A100", "A101", "A102"];
  return {
      tcp_ip: mockedData.join(",")
  };
}
```

Using Custom Templates

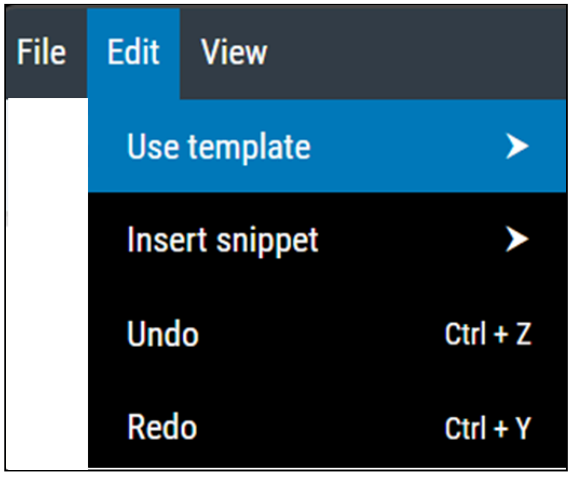
/docs/t-aurora-js-using-custom-templates

Using Custom Templates

Integrate your own JavaScript templates into Aurora Focus.

Create your template in a file with a `.js` extension, locate the Aurora Focus folder (where the application is installed), and save it in the `resources\\public\\js_editor_templates` subfolder.

After restarting the application, the template is available in the **Use template** section under **Edit** in the top navigation bar.



> **NOTE**
>
> The custom template you create is only available in that instance of Aurora Focus. If the file is deleted, the template is no longer available.

See Also

- Debugging, Advanced Techniques, and Sample Scripts

Monaco Sandbox

```
export function onResults(input_result) {
return {
    ftp: {
```

```
                common_file_name: "custom-template-output.txt"
            }
        };
        }
```

Obtaining the Device MAC Address

/docs/g-aurora-js-obtaining-the-device-mac-address

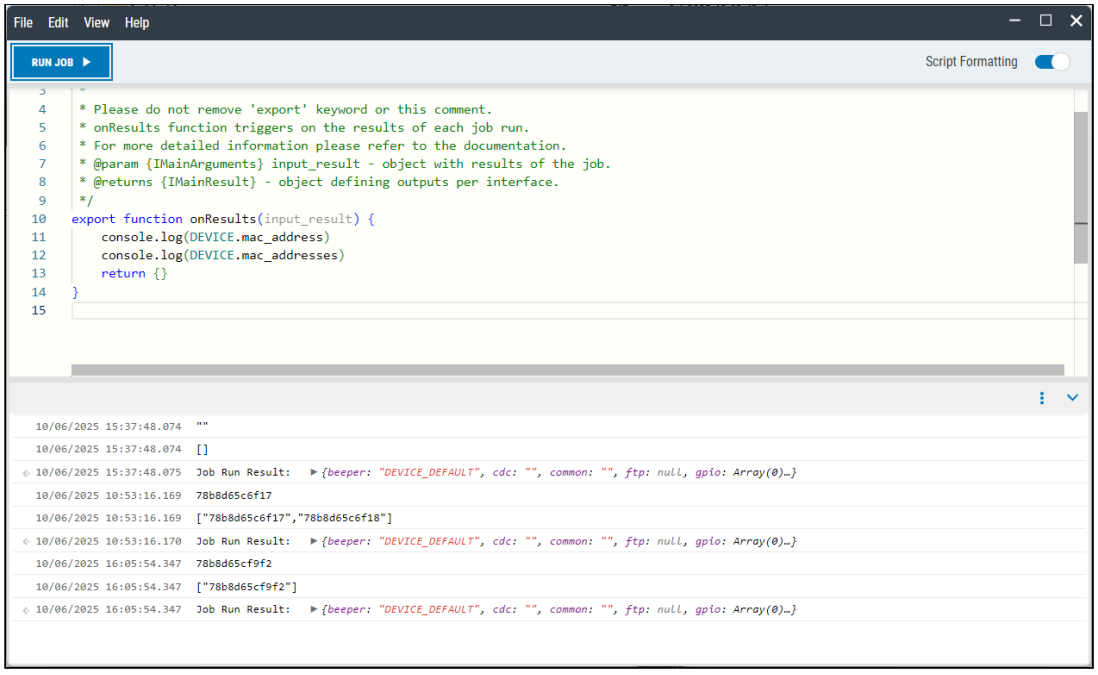
Obtaining the Device MAC Address

Use the Aurora Focus JavaScript functionality to obtain the device's MAC address.

Prerequisite

Use the following script to retrieve the device MAC address:

```
export function onResults (input_result) {
console.log(DEVICE.mac_address)
console.log(DEVICE.mac_addresses)
return {}
}
```



Sample Scripts

/docs/t-aurora-js-sample-scripts

Sample Scripts

The following sample scripts demonstrate different scenarios and editor features.

Click **Edit** on the top menu bar in the **Use template** section to use additional sample scripts. Each template has comments that describe the general purpose and blocks of code.

```
/**
* Read all codes using ManyCode, order them alphabetically, separate them
with a comma, and output them over TCP.
* Ensure that the Barcode Tool is in the first position in this job—
otherwise, the code might return errors and not work.
* Please do not remove the 'export' keyword or this comment.
* onResults function triggers on the results of each job run.
* For more detailed information, please refer to the documentation.
* @param {IMainArguments} input_result - object with results of the job.
* @returns {IMainResult} - object defining outputs per interface.
*/
export function onResults(input_result) {
  if (input_result.tool_results.decodes.length === 0) { // do not process
when there are no decodes
      return {};
  }
  const orderedDecodes = input_result.tool_results.decodes
      .map(decode => decode.value) // map array of decodes to array of
decoded values
      .sort() // sort alphabetically
      .join(','); // join elements of array with a comma to a single
string
  return {
      tcp_ip: orderedDecodes // output through tcp
  };
}

/**
* Capture barcodes with Manycode and compare two strings. If they're the
same output value, otherwise output "FAIL" through TCP and HID.
* Please do not remove the 'export' keyword or this comment.
* onResults function triggers on the results of each job run.
* For more detailed information, please refer to the documentation.
* @param {IMainArguments} input_result - object with results of the job.
* @returns {IMainResult} - object defining outputs per interface.
*/
export function onResults(input_result) {
  // return fail because there are no decodes to compare
  if (input_result.tool_results.decodes.length < 2) {
      return {
          job_success: false,
      };
  }

  // at this point, we are sure that there are at least two decodes here
due  to the check above
  // assign first and second decode value to variables for readability
  const firstDecode = input_result.tool_results.decodes.value;
  const secondDecode = input_result.tool_results.decodes.value;
```

```
  // compare decodes and return the decode value through TCP and HID if
they  are equal
  // set job_success to true, because the job did get two reads and did
run  correctly
  if (firstDecode === secondDecode) {
      return {
          tcp_ip: firstDecode,
          hid: firstDecode,
          job_success: true,
      };
  }

  // return "FAIL" because the first and second decode differ, and
job_success because the job did get two reads and did run correctly,
  // even though the decodes did not match
  return {
      tcp_ip: "FAIL",
      hid: "FAIL",
      job_success: true, // job success is true even though the decodes
do  not match - the job did run
  };
}
```

Monaco Sandbox

```
export function onResults(input_result) {
  return {
      tcp_ip: "Sample script sandbox"
  };
}
```